Appendix A Pseudo-code of ALNS components

This appendix provides the pseudo-code for three ALNS components summarized in Section 4 of the main paper.

A.1 Initial solution generation

Algorithm A1 below presents the greedy procedure used to generate the initial solution (referenced in Section 4.1).

Algorithm A1 Pseudo-code for generating initial solution

Require: instance $I = [V, V_D, V_C, \cdots, K, \cdots]$
Ensure: solution $S = [R, U, C]$, route set R, unvisited customer set U, total cost C

$v_c \leftarrow V_C$ // Customer set to be served
$k \leftarrow K$ // unused vehicle set
$k, r \leftarrow$ startNewRoute() // Start a new route, and remove the vehicle from unused vehicle set
while $|v_c| > 0$ **and** $|k| > 0$ **do**
    $c \leftarrow$ extendRoute(I, r) // Customer c is identified as the extension of the route
    if $c = \mathit{null}$ **then**
        finishRoute(r)
        $R \leftarrow R \cup \{r\}$ // Add the finished route to the route set
        $k, r \leftarrow$ startNewRoute()
    else
        $v_c \leftarrow$ remove(v_c, c) // Remove the customer for unvisited customer set
    end if
end while
finishRoute(r)
$R \leftarrow R \cup \{r\}$
if $|k| = 0$ **and** $|v_c| > 0$ **then**
    $U \leftarrow v_v$
end if
$R \leftarrow$ removeRedundantDepot(R) // Remove redundant depot nodes for each route
$C \leftarrow$ updateCost(C)

A.2 TP chain mechanism in repair operators

Algorithm A2 below details the TP chain mechanism used in the repair phase, with the random repair operator as the representative case (referenced in Section 4.3.1).

Algorithm A2 Pseudo-code for TP chain mechanism in RANDOM REPAIR

Require: solution S, routes R, customer c, insert position p
Ensure: new solution S'

$r \leftarrow \text{randomSelect}(R)$
$d \leftarrow \text{depot}(c)$
if d **not in** loadDepot(r) **then**
    $I \leftarrow \text{identifyTPInsertableInterval}(r, c, p)$
    $TP, cd_{min}, p_{TP} \leftarrow \text{minCost}(d, r, c)$
    $R' \leftarrow \text{potentialInteractionRoute}(R, d)$
    $C \leftarrow \text{identifyCandidateCustomerTP}(R', d)$
    while $C \neq \emptyset$ **do**
        $c' \leftarrow \text{removeFirst}(C)$
        $TP, cd_{min}, p_{TP} \leftarrow \text{minCost}(c', r, c)$
    end while
    if $TP = \text{null}$ **then**
        $S' \leftarrow \text{null}$
    else
        $r' \leftarrow \text{insert}(r, c, p, TP, p_{TP})$
        $S' \leftarrow \text{update}(S, r')$
    end if
end if

A.3 Depot-wise route matching for diversity-aware reheating

Algorithm A3 gives the depot-wise route matching procedure used to compute solution similarity in the diversity-aware reheating mechanism (referenced in Section 4.6).

Algorithm A3 Depot-wise route matching via the Hungarian algorithm

Require: two vehicle-slot-indexed route collections for depot d: $\mathcal{R}_d^c = (r_{d,1}^c, \ldots, r_{d,n_d}^c)$ and $\mathcal{R}_d^* = (r_{d,1}^*, \ldots, r_{d,n_d}^*)$

Ensure: optimal matching Π_d and total similarity Sim_d

for $i \leftarrow 1$ to n_d **do**
    for $j \leftarrow 1$ to n_d **do**
        $H_d[i,j] \leftarrow \textsc{RouteJaccard}(r_{d,i}^c, r_{d,j}^*)$
        $C_d[i,j] \leftarrow 1 - H_d[i,j]$
    end for
end for
$\pi_d \leftarrow \textsc{Hungarian}(C_d)$
$\Pi_d \leftarrow \emptyset$, $\mathrm{Sim}_d \leftarrow 0$
for $i \leftarrow 1$ to n_d **do**
    $j \leftarrow \pi_d(i)$
    $\Pi_d \leftarrow \Pi_d \cup \{(i,j)\}$
    $\mathrm{Sim}_d \leftarrow \mathrm{Sim}_d + H_d[i,j]$
end for
return Π_d, Sim_d

Appendix B Benchmark instances and parameter settings

B.1 Benchmark construction and instance summary

This subsection provides the benchmark construction details omitted from the main text. Following the naming convention ED-N-K, D, N, and K denote the numbers of depots, customers assigned to each depot, and trucks available at each depot, respectively. In all instances, each truck is paired with one robot, so the numbers of trucks and robots are identical. Robot-only customers account for 30% of the total customer population, and their number is computed as $\lceil 0.3DN \rceil$. Each robot-only customer is associated with two parking nodes. Therefore, the total number of nodes in an instance is given by

$$|V| = D(N+1) + 2\lceil 0.3DN \rceil. \tag{B1}$$

Appendix Table B1 provides a complete summary of the benchmark instances used in Section 5. For each instance, it reports the numbers of depots, customers per depot, total customers, trucks per depot, total trucks/robots, robot-only customers, and total nodes.

Table B1 Summary of benchmark instances used in Section 5.

Instances	Depots	Cust./depot	Total cust.	Trucks/depot	Total veh.	Robot-only cust.	Total nodes
E2-3-1	2	3	6	1	2	2	12
E2-4-1	2	4	8	1	2	3	16
E2-5-1	2	5	10	1	2	3	18
E2-6-1	2	6	12	1	2	4	22
E2-7-1	2	7	14	1	2	5	26
E2-8-1	2	8	16	1	2	5	28
E2-9-1	2	9	18	1	2	6	32
E2-10-1	2	10	20	1	2	6	34
E2-12-1	2	12	24	1	2	8	42
E2-16-2	2	16	32	2	4	10	54
E2-20-2	2	20	40	2	4	12	66
E2-30-2	2	30	60	2	4	18	98
E2-40-3	2	40	80	3	6	24	130
E2-50-4	2	50	100	4	8	30	162
E3-3-1	3	3	9	1	3	3	18
E3-4-1	3	4	12	1	3	4	23
E3-5-1	3	5	15	1	3	5	28
E3-6-1	3	6	18	1	3	6	33
E3-7-1	3	7	21	1	3	7	38
E3-9-1	3	9	27	1	3	9	48
E3-12-1	3	12	36	1	3	11	61
E3-15-1	3	15	45	1	3	14	76
E3-20-2	3	20	60	2	6	18	99
E3-30-2	3	30	90	2	6	27	147
E4-3-1	4	3	12	1	4	4	24
E4-4-1	4	4	16	1	4	5	30
E4-5-1	4	5	20	1	4	6	36
E4-6-1	4	6	24	1	4	8	44
E4-10-1	4	10	40	1	4	12	68
E4-15-1	4	15	60	1	4	18	100
E4-20-2	4	20	80	2	8	24	132
E4-25-2	4	25	100	2	8	30	164
E5-5-1	5	5	25	1	5	8	46
E5-10-1	5	10	50	1	5	15	85
E5-15-1	5	15	75	1	5	23	126
E5-20-2	5	20	100	2	10	30	165

Note: The columns Depots, Cust./depot, and Trucks/depot correspond to D, N, and K in the instance naming convention, respectively. Total cust. denotes the total number of customers (DN). Total trucks/robots denotes the total numbers of trucks and robots, which are identical and equal to DK, since each truck is equipped with one robot. Robot-only cust. denotes the number of robot-only customers, computed as $\lceil 0.3DN \rceil$. Total nodes denotes the total number of nodes, computed as $D(N+1) + 2\lceil 0.3DN \rceil$.

B.2 Default instance parameters

Appendix Table B2 presents the default parameters used for instance generation. The instance region is a square area of 100×100 km, within which D depots are discretely located away from both the edges and the center. All customers are distributed within this region. For each robot-only customer, the distances to its two associated parking nodes follow a continuous uniform distribution $U(1, 5)$ km (Yu et al. 2022). Customer demands follow a uniform distribution $U(10, 30)$. The fixed cost for truck utilization, denoted by c_f, is set to 50. The truck travel cost c_v is 2/km, and the truck load capacity U_t is 300 kg (Chen et al. 2021). The maximum travel distance $\mathscr{D}$ for a single truck trip is limited to 300 km. The robot travel cost c_r is 1/km, and the robot load capacity U_r is 30 kg (Simoni et al. 2020). The robot energy consumption ε is 0.1 kWh/km, and the battery capacity $\mathscr{E}$ is 3 kWh.

Table B2 Default instance parameters

Symbols	Values	Unit
c_v	2	/km
c_r	1	/km
c_f	50	–
U_t	300	kg
U_r	30	kg
ε	0.1	kWh/km
$\mathscr{E}$	3	kWh
$\mathscr{D}$	300	km

B.3 ALNS configuration

Appendix Table B3 reports the default parameter settings of the ALNS algorithm. These settings are used throughout the computational experiments unless otherwise specified.

Table B3 ALNS configuration

Parameters	Symbols	Values
Weight adjustment coefficient	λ	0.5
Score of the repaired solution is superior to the best solution	σ_1	10
Score of the repaired solution is superior to the current solution	σ_2	6
Score when the simulated annealing criterion is met	σ_3	1
Maximum number of iterations	–	100000
Consecutive iterations for weight updates	ϑ_s	300
Maximum number of consecutive iterations without updates	–	10000
Temperature cooling rate	α	0.9985
Capacity of solution pool	–	200
Base reheating factor	ρ	1.1
Threshold of similarity	θ	0.5
Removal percentage	p	30%
Low-temperature threshold ratio	η	0.1

Appendix C Detailed computational results

C.1 Instance-level results for the diversity-aware reheating comparison

Appendix Table C4 reports the complete instance-level results of the comparison between the standard ALNS and the enhanced ALNS summarized in Section 5.2.2. For each of the 21 larger-scale instances, the table lists the best objective value, the average objective value, and the average runtime over five runs obtained by each algorithm, together with the corresponding relative changes of the enhanced ALNS with respect to the standard ALNS.

Table C4 Instance-level comparison of standard ALNS and enhanced ALNS with diversity-aware reheating on larger-scale instances.

	Standard ALNS			Enhanced ALNS			Relative change (%)		
Instances	Best obj.	Avg obj.	Avg T. (s)	Best obj.	Avg obj.	Avg T. (s)	ΔBest obj.	ΔAvg obj.	ΔAvg T.
E2-12-1	1157.86	1157.86	50.20	1157.86	1157.86	50.55	0.00	0.00	0.70
E2-16-2	1289.78	1296.19	67.83	1289.78	1295.02	128.49	0.00	-0.09	89.43
E2-20-2	1490.56	1505.58	148.27	1478.22	1486.34	210.63	-0.83	-1.28	42.06
E2-30-2	1865.17	1901.85	198.30	1854.58	1876.70	235.79	-0.57	-1.32	18.91
E2-40-3	2528.66	2586.27	449.80	2483.70	2545.69	477.54	-1.78	-1.57	6.17
E2-50-4	2597.68	2705.02	1914.73	2598.62	2655.31	1687.53	0.04	-1.84	-11.87
E3-7-1	1192.84	1198.67	29.02	1192.84	1197.26	32.56	0.00	-0.12	12.20
E3-9-1	1222.26	1237.47	61.08	1222.26	1227.64	66.44	0.00	-0.79	8.77
E3-12-1	1533.66	1555.79	102.54	1526.04	1546.89	116.69	-0.50	-0.57	13.80
E3-15-1	1670.17	1687.75	275.71	1630.10	1674.50	486.98	-2.40	-0.79	76.63
E3-20-2	1891.77	1966.39	714.38	1845.98	1875.34	1024.70	-2.42	-4.63	43.44
E3-30-2	2587.69	2662.64	1586.01	2473.30	2520.23	2003.04	-4.42	-5.35	26.29
E4-6-1	1254.90	1271.88	33.15	1254.90	1267.86	51.76	0.00	-0.32	56.14
E4-10-1	1599.00	1631.89	123.03	1585.89	1628.64	150.65	-0.82	-0.20	22.45
E4-15-1	1919.34	1947.38	400.12	1846.82	1941.65	301.87	-3.78	-0.29	-24.56
E4-20-2	2438.09	2472.95	1649.96	2396.48	2446.64	2248.16	-1.71	-1.06	36.26
E4-25-2	2855.87	2896.56	1716.24	2787.79	2833.97	1800.35	-2.38	-2.16	4.90
E5-5-1	1535.04	1586.93	59.19	1487.76	1551.90	63.63	-3.08	-2.21	7.50
E5-10-1	1908.84	1963.23	648.65	1866.80	1945.07	566.51	-2.20	-0.93	-12.66
E5-15-1	2370.60	2451.76	747.47	2364.21	2396.61	811.37	-0.27	-2.25	8.55
E5-20-2	2997.10	3051.39	3334.24	2885.81	3013.85	2213.68	-3.71	-1.23	-33.61
Mean relative change							-1.47	-1.38	18.64

Note: ΔBest obj., ΔAvg obj., and ΔAvg T. denote the relative changes of the enhanced ALNS with respect to the standard ALNS. The last row reports the mean relative changes over all 21 instances.

C.2 Instance-level results for the no-horizontal-collaboration variant

Appendix Table C5 reports the complete instance-level results of the no-horizontal-collaboration (NH) variant summarized in Section 5.2.3. For each benchmark instance, the table lists the average objective value obtained by the NH variant and its relative change in Avg obj. with respect to the corresponding baseline result.

Table C5 Computational results of the no-horizontal-collaboration (NH) variant relative to the baseline HMDVRRP model.

Instance	Avg obj.	ΔAvg obj.	Instance	Avg obj.	ΔAvg obj.
E2-3-1	835.38	67.63%	E3-7-1	1721.10	44.29%
E2-4-1	865.96	63.91%	E3-9-1	–	–
E2-5-1	1009.56	18.11%	E3-12-1	–	–
E2-6-1	1072.60	24.29%	E3-15-1	–	–
E2-7-1	1200.98	22.31%	E3-20-2	2918.89	58.12%
E2-8-1	1222.16	20.91%	E3-30-2	–	–
E2-9-1	1282.58	21.15%	E4-3-1	1171.70	29.11%
E2-10-1	1254.22	16.62%	E4-4-1	1627.26	47.19%
E2-12-1	–	–	E4-5-1	1883.62	52.93%
E2-16-2	1911.60	48.21%	E4-6-1	2066.66	64.69%
E2-20-2	1989.04	34.56%	E4-10-1	2474.11	56.01%
E2-30-2	–	–	E4-15-1	–	–
E2-40-3	2905.20	16.97%	E4-20-2	3736.41	55.91%
E2-50-4	3466.47	33.40%	E4-25-2	4248.79	52.41%
E3-3-1	1013.52	59.51%	E5-5-1	2460.06	65.35%
E3-4-1	1070.42	26.74%	E5-10-1	–	–
E3-5-1	1303.42	28.34%	E5-15-1	–	–
E3-6-1	1646.62	45.64%	E5-20-2	4996.18	73.13%

Note: Avg obj. denotes the average objective value obtained by the NH variant. ΔAvg obj. denotes the relative deviation from the corresponding baseline result. A dash indicates that the NH variant is infeasible for the corresponding instance.